

ECO209 Writing Assignment 2

2024-2025

Professor Yip

Due date: Mar 30th, 2025

Background

Japan experienced remarkable economic growth following World War II. However, beginning in the early to mid-1990s, the country entered a prolonged period of sluggish economic expansion, commonly referred to as Japan's Lost Decades. Multiple factors contributed to this economic downturn. The objective of this assignment is to analyze Japan's Lost Decades and derive lessons that could help Canada avoid a similar situation.

Instructions

As a junior economist for the Canadian government, you have been assigned to prepare a background report on Japan's Lost Decades and extract lessons to inform policy recommendations.

This report is structured into three sections:

- 1) Analyze Japan's real GDP by examining its trend and cyclical components in comparison to an assigned country.
- 2) Identify a specific cause of Japan's Lost Decades and provide a background summary.
- 3) Based on the findings from Parts 1 and 2, propose a policy recommendation.

Your report will go to the ministers' office, all of whom have a moderate understanding of intermediate macroeconomic theory. You may assume this audience has taken ECO209 in the past and can follow descriptions of our economic model. You may reference model terms and variables *as long as* you properly define them. For example, **do not** simply say 'this will cause an increase in the Marginal Product of Labour'. Instead, say 'this will cause an increase in the Marginal Product of Labour (MPL), which tells how much additional output when a firm increases its labour input'. Wherever possible, use English terms to refer to model terms rather than mathematical terms/formulas.

Your report must have a **minimum** of two data plot. All plots must contain an accurate title, a legend for every data series, and labels on all axes. You may include the plots either in the body of the text or in an appendix at the end of the report. Ensure each plot is labelled with a convenient name for referencing in the text, i.e., 'Figure 1'. Extra plots can be included if it assists you in your argument in the essay.

Your division has strict formatting rules for this report that you must follow. The report must:

- Include a **brief** introduction that should provide a broad background for Japan's Lost Decades in one to two paragraph (part 0).
- Be **typed, double-spaced**, and have a **font size of 12 points**.
- Be between **800-1400 words long**. This is a strict cut-off, and you will lose points for writing more than 1400 words. Put the word count of your assignment in parentheses at the end, like "(943 words)."
- **Include a header** (the first four lines) with your full name and student ID number, the course information (ECO209, 2024–2025), and the title of your assignment.
- **Cite the data sources**. You can simply use the suggested citation from the FRED website. You do not need to cite the lectures, class notes, or the textbook. However, if you use any other articles (including the optional background paper) you must cite them. Any reference format is accepted but

they must be consistent (e.g., if you use APA, it must be APA throughout except for the data citation). All references should be listed at the end of the paper. This reference list does not count towards the final word limit.

Assignment Details

Part 0: Introduction

Provide a broad background for Japan's Lost Decades and give an overview of your analysis in Part 1 in one to two paragraphs.

Here is a good starting point, an opinion piece by Paul Krugman in 2015:

<https://archive.nytimes.com/krugman.blogs.nytimes.com/2015/10/20/rethinking-japan/>

Part 1:

First, download the Constant GDP per capita (in 2010 USD)¹ from FRED, Federal Reserve Bank of St. Louis (<https://fred.stlouisfed.org>) from 1970 to 2023 for Japan and your randomly assigned country (see the "*List of randomly assigned country.xlsx*" on Quercus). The frequency of all data series should be annual. To make sure you have the right data series, you should use the list of variable labels provided on Quercus (*Data Label FRED.xlsx*).² Record the citation for each data series as a reference.

Second, calculate the potential output (trend) and the short-run output (percentage deviation from the trend) using the Hamilton filter. The Hamilton filter regression is estimated separately for each country.

Japan:

$$y_{t+4} = 4535.5031 + 0.9289y_t - 0.1583y_{t-1} + 0.1247y_{t-2} + v_{t+4}$$

US:

$$y_{t+4} = 2638.4409 + 1.0797y_t - 0.5816y_{t-1} + 0.5165y_{t-2} + v_{t+4}$$

Canada:

$$y_{t+4} = 4026.8753 + 1.0576y_t - 0.4043y_{t-1} + 0.2806y_{t-2} + v_{t+4}$$

France:

$$y_{t+4} = 4221.5512 + 0.8965y_t - 0.3937y_{t-1} + 0.4130y_{t-2} + v_{t+4}$$

Singapore:

$$y_{t+4} = 2604.9539 + 0.8266y_t - 0.3043y_{t-1} + 0.5685y_{t-2} + v_{t+4}$$

Netherland:

$$y_{t+4} = 3175.6631 + 1.3818y_t - 0.8271y_{t-1} + 0.4197y_{t-2} + v_{t+4}$$

Demark:

$$y_{t+4} = 4267.0171 + 1.0233y_t - 0.3082y_{t-1} + 0.2482y_{t-2} + v_{t+4}$$

¹ It is just another name of Real GDP.

² Hints: (1) You can search the code on the FRED website for the data (2) Use ``Edit Graph" --> ``Add Line" to add another data series on the same sheet

Third, compare the potential output and short-run output between Japan and the randomly assigned country. Discuss the differences observed in Japan during the 1980s, the 1990s, and the periods that followed.

In your writing, include two to three sentences explaining the method used to separate the trend and cyclical components of real GDP per capita in a simple way. Keep in mind that your target audience is not technically inclined, so avoid complex explanation and terminology.

Part 2: Highlight One of the causes of the Japan's Lost Decades

Select and conduct research on *one* of the causes of Japan's Lost Decades. Your interpretation should be based on either the Growth Model (i.e., focusing on potential output) or the IS-MP model (i.e., focusing on current output). To support your argument, you may include citations and use additional data. Your discussion should be connected to the findings from Part 1.

Part 3: Lessons Learned and Policy Recommendations

Finally, derive a key lesson from Part 2 and propose a policy recommendation for the Canadian government to help prevent Canada stepping into the same sluggish economic growth situation. If relevant, highlight any parallels between challenges faced by Canada today and those experienced by Japan.

Part 0 to 3 should be written in a clear, and well-connected manner. A good paper is the one that each paragraph flows logically from the previous one, maintaining smooth transitions and connected throughout. Each paragraph should focus on a single main idea, presenting it in a structured way that effectively explains and supports the argument.

When you reference a data plot you must describe the variables it contains, and the trend shown by the data. Do not assume that the reader will look at the graphs and independently understand their meaning. You must guide your audience towards your conclusions within the body of the text.

Submission instructions

You must submit your assignment to Quercus. To do so, follow these steps:

- 1) Complete your essay in Microsoft Word or other type writing program (but you must export it as word file (.doc or .docx file) when submitting.
- 2) Submit the Microsoft Word file and the spreadsheet file to the Quercus assignment. You may upload a pdf file as a backup if you are exporting to Word file)
- 3) Must be completed before the deadline on Mar 30th at 11:59pm!**

You must verify your work was properly submitted before the deadline. You may make multiple submissions before the deadline if you discover an issue with a previous submission.

*Refer to the separate rubric for details on how your work will be evaluated.